



國立陽明交通大學資訊工程學系資訊中心

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HW1-3 - LDAP

2026 NAP
ymlai

Objectives

- Build a basic LDAP service
- Understand how to...
 - Configure an LDAP server
 - Manage LDAP data using LDIF
 - Perform authentication and permission control on a Unix client with an LDAP server
 - Customize your own object classes and use OLC (online configuration)

Outline & Grading

- Basic Configuration (0%)
- Organizational Unit Naming (0%)
- PosixGroup (4%)
- People (5%)
- Access control (8%)
- Password policy (7%)
- TOTP (6%)
- Fortune (4%)

Basic Configuration

- DNS Record
 - ldap server(private zone): `ldap.${STUID}.nasa`
 - workstation1(DMZ): `workstation1.${STUID}.nasa`
 - workstation2(DMZ): `workstation2.${STUID}.nasa`
 - The `{STUID}` is same as the ID you use in HW1-1
- Base DN: `dc=<STUID>, dc=nasa`
- LDAPS and force TLS search
 - Not LDAP over TLS (StartTLS)
 - Generate your key and certificate by yourselves

Organizational Unit Naming

- People
- Group (posixGroup)
- Ppolicy
- SUDOers
- Fortune (HW customize objectClass)

PosixGroup

We need two posix groups in LDAP:

- **ta** group (GID=10000)
 - Can login (ssh) into all servers and all workstations
 - Can use sudo for **any command**
 - ex. ``sudo adduser``
- **stu** group (GID=20000)
 - Can login (ssh) into workstation1 only
 - Only allow sudo for ``ls`` command
- Please **use LDAP** to implement the requirements above.
 - Including sudo rules and ssh key!

People (1/3)

Add a user with DN “uid=generalta,ou=People,<Base DN>”

- This user under **ta** group, use **ta** group permission
- Allow this user to connect via SSH with both ssh public key and password
 - uid: generalta
 - uid number: 10000
 - homeDirectory: /u/ta/generalta
 - public key: ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMUa1AsYerXm5/QVrTZ7QxtkcCUfuXopO04xu2hOBxNY 2026 NAP
 - user password: <TA_PASSWORD>
 - user's password must be hashed
 - mail: generalta@{STUID}.nasa

People (2/3)

Add a user with DN “uid=mailta,ou=People,<Base DN>”

- This user under **ta** group, use **ta** group permission
- Allow this user to connect via SSH with both ssh public key and password
 - uid: mailta
 - uid number: 10001
 - homeDirectory: /u/ta/mailta
 - public key: ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMUa1AsYerXm5/QVrTZ7QxtkcCUfuXopO04xu2hOBxNY 2026 NAP
 - user password: <TA_PASSWORD>
 - user's password must be hashed
 - mail: mailta@{STUID}.nasa

People (3/3)

Add a user with DN “uid=stu, ou=People, <Base DN>”

- This user under **stu** group, use **stu** group permission
- Allow this user to connect via SSH with both ssh public key and password
 - uid: stu
 - uid number: 20000
 - homeDirectory: /u/stu/stu
 - public key: ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMUa1AsYerXm5/QVrTZ7QxtkcCUfuXopO04xu2hOBxNY 2026 NAP
 - user password: <TA_PASSWORD>
 - user's password must be hashed
 - mail: stu@{STUID}.nasa

Access Control (1/2)

- Allow **generalta** and **mailta** to manage users and groups
- Allow all users to modify their own
 - `userPassword`, `loginShell` and `sshPublicKey`
 - Set all other attributes as **read-only**
- Allow users to search all user data, except for other users' password
 - i.e., users can only read their own password
 - **generalta** can write passwords but must not be allowed to read them!!

Access Control (2/2)

- Mount the home directory on each ldap clients
 - Path: /u/ta, Permission: 755, Owner: root:root
 - Path: /u/ta/{ta_name}, Permission: 711, Owner: {ta_name}:ta
 - Path: /u/stu, Permission: 755, Owner: root:root
 - Path: /u/stu/{stu_name}, Permission: 711, Owner: {stu_name}:stu
- Note:
 - Home directory files for both students (stu) and teaching assistants (ta) must be **consistently synchronized** across all machines

Password policy

- userPassword can't be same as previous when change password
 - But can set password as previous two time used
 - You need to implement this by LDAP way
- Password requires **at least 8** characters long
- Password must contain **at least 3** different classes of characters:
 - Upper-case characters
 - Lower-case characters
 - Digits
 - Special characters

Time-based one-time passwords (TOTP)

- Apply TOTP feature on **general**ta and **stu** account
 - TOTP authentication is required **only for** password-based logins
 - Logins using SSH public key authentication are exempt from TOTP authentication
 - The OJ will test authentication by requiring the password followed by the TOTP code
 - For example: secret123456
- Add the TOTP schema to the LDAP server
 - oathOTPLength: 6
 - oathTOTPTimeStepPeriod: 60
 - oathHMACAlgorithm: SHA-1
 - oathSecret: <TOTP_secret>

Fortune (1/2)

Add an ObjectClass fortune with on-line configuration(OLC)

- **schema**

- ObjectClass's oid should be under the [UUID branch](#)
 - Extend from **top**, add **author** field(Directory String), and **id** field(integer)
 - **author's** matching, substring and order should be “**case insensitive, space insensitive**”
 - Use existing **description**([RFC 4519](#)) attribute to place sentences
- You need to check whether this objectclass is in database (cn=config)

Fortune (2/2)

- Import fortunes
 - From given yaml file ([link](#))
 - 3 fields
 - ID
 - Author
 - Description
- Enable features
 - Server side sorting
 - Pagination
 - Hint: slapd-sssylv

```
ID: 106
Author: Richard Feynman
Description: 'I don''t know what''s the matter with people: they don''t learn by understandin
```

```
dn: cn=fortune-1,ou=Fortune,dc=254,dc=nasa
objectClass: fortune
objectClass: top
cn: fortune-1
author: Richard Feynman
id: 1
description: The first principle is that
you must not fool yourself -- and you are
the easiest person to fool.
```

LDAP Client

- Configure LDAP Client on workstation1, workstation2
- Configure LDAP for login (ssh) authentication
 - LDAP user can use password or public key to login
 - Login permissions at [PosixGroup](#)

Note

- You need to generate the **TA_PASSWORD** and **TOTP_secret** in
 - OJ → Tools → “Generate TA password & TOTP secret”
- OJ will use the **generalta** account to test all requirements in hw1-3
 - The **mailta** account will be used to test requirements related to LDAP in hw1-2
- **Always backup** your system before submission, as we may perform malicious actions during testing
- Make sure everything works correctly after **reboot**

Good Luck!